Kuanwei Chen

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Summary

M.S. Computer Science student at National Central University specializing in Artificial Intelligence, with research presented at three international conferences. Experienced in building computer vision and NLP systems for scene text recognition, sign language translation, and biometric recognition using PyTorch and TensorFlow. Seeking AI/ML engineering or computer vision research roles.

Education

National Central University, Taoyuan, Taiwan *Sep 2025 – Present*
Master in Computer Science, specialization in Artificial Intelligence

National Changhua University of Education, Changhua, Taiwan *Sep 2021 – Jun 2025*
Bachelor in Electrical Engineering, GPA 3.8/4.0

Experience

Project Assistant in AI Application Laboratory *Jun 2024 – Dec 2024*

- Collaborated with 8 international interns under the TEEP and IIPP research programs, delivering 3 multimodal AI prototypes in biometric and action recognition using PyTorch and TensorFlow, contributing to a peer-reviewed publication at IET ICETA.
- Built Pandas and NumPy-based preprocessing, dataset, and evaluation pipelines that streamlined model experimentation and validation across computer vision and speech-related research tasks.

Publication

STRLite: Lightweight Masked Autoencoders for Adaptable Scene Text Recognition
IEEE ICCE-TW | github.com/balaboom123/STRLite

- Co-first author; designed a 6M-parameter Vision Transformer (ViT) with Masked Autoencoder (MAE) pretraining and fine-tuning with an autoregressive decoder for Scene Text Recognition.
- Achieved 93.82% weighted accuracy on six standard Computer Vision benchmarks and 81.03% on U14M.

Towards Compact Sign Language Translation: Frame Rate and Model Size Trade-offs
IEEE ICCE-TW | arXiv:2605.09554

- First author; built a 77M-parameter end-to-end Natural Language Processing (NLP) pipeline.
- Reduced computation by ~75% with minimal BLEU impact (10.06 to 9.53), providing efficiency trade-off insights.

Two-step Authentication: Multi-biometric System Using Voice and Facial Recognition
IET ICETA | doi.org/10.1049/icp.2024.4141 | github.com/NCUE-EE-AIAL/Two-Step-Muti-Biometric-Authentication-System

- First author; led a team of 2 interns in designing a multi-modal AI/ML system combining Machine Vision and Audio Recognition/Speaker Verification.
- Designed a ResNet-based Convolutional Neural Network (CNN) with Fbank feature extraction and triplet loss for voice verification (99.1%); fine-tuned VGG16 with MTCNN object detection in TensorFlow (95.1%).

Projects

SignDATA: Data Pipeline for Sign Language Translation
arXiv:2604.20357 | github.com/balaboom123/signdata-slt

- First author; open-source repo with 100+ stars; built a config-driven data preprocessing, and dataset management system for standardizing heterogeneous sign-language corpora for AI/ML Pipeline development.
- Designed processing pipeline integrating YOLO/MMDet detection and MediaPipe/MMPose landmark extraction for structured feature engineering, applying system-design principles for multi-dataset scalability.

SEO Content Intelligence Engine
github.com/balaboom123/SEO-Content-Intelligence-Engine | seo-content-intelligence-engine.vercel.app

- Built a full-stack Next.js web app in TypeScript that scrapes top-10 Google SERP results, extracts NLP entities via OpenAI, and clusters them to reverse-engineer search intent for SEO competitor analysis.
- Implemented Supabase authentication with SQL Row-Level Security policies, interactive Recharts visualizations, Google Sheets export, and deployed on Vercel with rate limiting and API security hardening.

Skills

Programming & Tools	Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Git, Linux, Docker
ML Frameworks & Stack	PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV, NumPy, Pandas, YOLO, MMDet, MediaPipe
Web Development	React, Next.js, Vue, Supabase, Vercel

Languages: English (IELTS 6.5, TOEIC 830), Mandarin (native), Hokkien (native)